

## EJAY AGUIRRE

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### EDUCATION

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#### Kahlert School of Computing, University of Utah

Bachelor of Science in Data Science, Minor in Environmental Sustainability Studies  
Generation Next Scholar, NASA Space Grant Consortium Scholar, WICHE Recipient

Salt Lake City, UT  
Expected May 2028

#### Singapore Management University

Benjamin A. Gilman International Scholarship Recipient, Data Management with SQL

Singapore  
Summer 2025

#### Northern Marianas College

Associate Arts of Liberal Arts Emphasis in Pre-Engineering, Magna Cum Laude Honors  
NFTA-APIA Scholar, NextGen Scholar, Western Digital Scholar, New Century Transfer Pathway Scholar

Saipan, MP  
May 2022

### SKILLS

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**Technical:** Python, R, SQL, Jupyter, Java, Git, AutoCAD, MATLAB, Data Pipelines (ETL), APIs, GIS

**Data Science:** Data cleaning & preprocessing, time-series & statistical analysis, reproducible workflows

**General:** Computer Proficient, Project Management, Communication, Time Management, Research

### EMPLOYMENT

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#### Neuroergonomics and Occupational Biomechanics Laboratory

Research Assistant

Salt Lake City, UT  
April 2025 - Present

- Proposed an independent research project on occupational noise, muscle fatigue, and **EMG** analysis that was accepted for funding, supporting the lab's broader work on worker safety and ergonomics.
- Collected and supported experimental data acquisition using **Biosemi** and **E-Prime** systems to support ongoing research projects from senior researchers.
- Analyzed EMG and biomechanical data to investigate occupational ergonomics and human performance using **MATLAB** and the **EEGLab** Library.
- Presented early findings at the University of Utah Undergraduate Research Symposium and the Rocky Mountain American Society of Biomechanics regional meeting.

#### Wasatch Community Gardens

GIS Intern

Salt Lake City, UT  
May 2026 - July 2026

- Built a GIS asset management system to map irrigation, garden beds, utilities, and infrastructure across four community garden sites using **ArcGIS Pro**, **ArcGIS Online**, and **Field Maps**.
- Designed a 12-layer **geodatabase** with coded-value domains, standardized field forms, editor tracking, photo attachments, and automated geometry calculations to improve field-data consistency.
- Collect, validate, and visualize **GPS** and infrastructure data to produce public-facing and technical maps supporting site operations, irrigation maintenance, and long-term planning.

#### University of Utah Health

IT Support Technician

Salt Lake City, UT  
October 2025 - March 2026

- Supported **SharePoint Online** migration workflows, including site provisioning, permissions mapping, and data transfers, ensuring data integrity through validation checks.
- Triaged IT tickets and walked users through **SharePoint/OneDrive** basics (access, syncing, versioning), escalating cases, and documenting fixes to build repeatable workflows.
- Documented and standardized recurring technical issues and resolutions, improving troubleshooting efficiency and creating reusable workflows for system support.

#### Ocean Observatories Initiative

Data Analyst Intern

Woods Hole, MA  
June 2025 - August 2025

- Implemented quality control across ETL workflows in **Python** to verify data integrity, catching inconsistencies in time-series data pulled via **ERDDAP** and **APIs** and improving reliability.

- Developed **Python** scripts and **Jupyter Notebook** pipelines to retrieve, clean, and aggregate ~12 years of in-situ water temperature and CTD data via ERDDAP, streamlining time-series data.
- Created publication-quality plots in **Matplotlib** & **ggplot**, overlaying ENSO phases on temperature trends across multiple stations, highlighting the climatic phenomena and accessibility to data.
- Authored comprehensive documentation of the **ETL-style** notebooks, ensuring reproducibility and handoff readiness for interdisciplinary research teams.

## CI Compass

Salt Lake City, UT

Student Fellow

January 2025 - April 2025

- Engaged in a 12-week program focusing on technical and research skills relevant to cyberinfrastructure and data-intensive research.
- Gained a contextual understanding of Major Facilities and the data lifecycle, enhancing the ability to support collaboration and inquiry.
- Collaborated with peers and mentors to apply learned skills in real-world scenarios, preparing for future roles in cyberinfrastructure and scientific computing.

## Center for Student Access & Resources

Salt Lake City, UT

Peer Mentor

September 2024 - April 2025

- Mentored undergraduate students through personalized academic planning, time management strategies, and guidance on utilizing university resources, fostering academic success and healthy relationships with peers and professors.
- Organized peer group meetings, socials, and workshops in collaboration with staff, cultivating a supportive community and increasing student engagement in campus programs.

## Salt Lake City Corporation

Salt Lake City, UT

Constituent Liaison Intern

September 2024 - January 2025

- Managed constituent inquiries and concerns using **Salesforce** to track case statuses, streamline communication between city officials and the public, and ensure timely resolutions, improving community satisfaction.
- Prepared detailed reports and case studies for city officials by utilizing **Microsoft Office** tools to analyze data, summarize local issues, and propose actionable solutions.

## C.O.M.P.A.S.S | alternative REU

Saipan, MP

Data Researcher Lead Intern

June 2024 - August 2024

- Designed and implemented a green roof model analyzing soil composition, moisture retention, and evapotranspiration, supporting stormwater management research for rural development.
- Conducted correlation and t-test **analyses** in **Python**, interpreting results to inform actionable conclusions; authored a research chapter documenting the project for future publication.

## University Centers for Excellence in Developmental Disabilities

Hawaii, HI

Student Trainee

June 2023 - January 2024

- Evaluated the educational and technological needs of individuals with disabilities, ensuring compliance with cultural competence, ethical standards, and developmental disability policies.
- Collaborated with professionals and advocates to identify and implement solutions, enhancing knowledge of cross-disciplinary communication and inclusivity best practices.

## S.P.I.C.E Data Science Institute

Hawaii, HI

Data Researcher Intern,

May 2023 - July 2023

- Analyzed datasets on avian elevation and malaria correlations, creating visualizations with graphic design tools to effectively communicate findings to professionals and cohort members.
- Utilized **R** and Texas Advanced Community Centers (**TACC**) software to process research data, strengthening technical expertise and demonstrating the ecological impact of bird elevation on avian malaria infection rates.

## PROJECTS

### Textbook-to-Podcast Pipeline

- Built an end-to-end **AI** pipeline that parses textbook PDFs by chapter, generates conversational two-host podcast scripts via **LLM** prompt engineering, and synthesizes dual-voice audio using **ElevenLabs TTS**
- Implemented a semantic hallucination detection system using sentence-transformers to score AI-generated scripts against source material, flagging content drift with cosine similarity thresholds
- Engineered a modular **CLI** tool with argparse supporting chapter filtering, **TTS** toggling, and configurable quality thresholds across a 4-stage processing pipeline

### Green Roof Opportunity Screening

- Built a reproducible pipeline to rank building footprints for green roof opportunity using Annual NLCD Fractional Impervious Surface, Land Cover, and Tree Canopy, utilizing **Python's** geospatial library, which is shared on a **GitHub** repository.
- Computed per-building zonal statistics and produced a stakeholder-facing GeoJSON layer with transparent scoring, tier labels, and QA flags utilizing **ArcGIS Online**, which was published as an interactive web map and in a narrative **StoryMap**.

### GeoPulse - Data Analytics & Integration Project

- Built a **full-stack** geospatial dashboard that processes Landsat 8/9 satellite imagery via **GEE** to identify optimal geothermal development sites across Utah
- Engineered a pipeline using the Sobrino (2004) single-channel method to derive land surface temperature from raw thermal satellite bands
- Designed a weighted scoring algorithm ( $GPS = 0.5 \times LST + 0.3 \times Grid + 0.2 \times SVI$ ) integrating the CDC Social Vulnerability Index 2022 to prioritize energy-underserved communities

## LEADERSHIP & ACTIVITIES

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### Phi Theta Kappa Beta Lambda Psi Chapter

Saipan, MP

Public Relations Officer | HIA Lead

October 2023 - May 2024

- Led research efforts to create visually impactful representations of cultural engagement using qualitative analysis and technological tools, earning recognition from the International Society for innovative content creation.
- Mentored students with disabilities by sharing research strategies and offering guidance, resulting in successful collaboration and a joint student-mentor presentation at the Mariana Islands Conservation Conference.